Aman Kumar Bhamboo

amankumarbhamboo14022005@gmail.com · +91 6367755376 · LinkedIn · GitHub · Tableau

SUMMARY

Final year CS student with proven experience building production grade ETL pipelines, SQL cohort analytics, and multi-page Tableau dashboards on datasets exceeding 1GB. IEEE-published researcher who turns fragmented raw data into boardroom-ready insights that drive measurable business decisions.

ACHIEVEMENTS

Research Publications

"Deep Learning-Based Approaches for Efficient Brain MRI Segmentation: A Comprehensive Analysis," IEEE STCR 2025, pp. 1-9, doi: 10.1109/STCR62650.2025.11020327.

- Aggregated MRI datasets across multiple hospital sources and benchmarked U-Net, ResNet, and transformer-based architectures for brain MRI segmentation across accuracy, computational efficiency, and clinical applicability in resource-constrained environments.

PROJECTS

LAPD Crime Analysis — End-to-End Data Pipeline & Insights — GitHub | Tableau Dashboard

- **Problem:** Raw LAPD crime records spanning 14 years were fragmented across inconsistent schemas, making trend analysis and public safety insights nearly impossible at scale.
- Identified geographic crime hotspots via LAT/LON spatial clustering, revealing **top 5 high-incidence divisions** and supporting location-based resource allocation decisions.
- Conducted large-scale EDA uncovering **peak crime hours (10 PM–2 AM), year-over-year clearance rate decline post-2015,** and weapon involvement patterns across 28 crime attributes with documented null analysis justifying column exclusions.
- Engineered a config-driven Python ETL pipeline (ingest → transform → merge → PostgreSQL export) processing 3.1M records (∼1 GB) with YAML-driven schema config, custom exception handling, and dual-channel loggingcutting data prep from hours to under 6 minutes.
- Built an interactive Tableau dashboard visualizing crime trends, hotspot maps, demographic distributions, and clearance metrics enabling non-technical stakeholders to explore **14 years of public safety data** without SQL.

Measuring Customer Revenue Health — GitHub | Tableau Dashboard

- **Problem:** A Brazilian e-commerce marketplace had zero visibility into why customers weren't returning with ∼100K orders and no retention or churn metrics in place.
- Uncovered an **8% repeat purchase rate** and **77% CSAT**, with cohort analysis pinpointing >**70% churn within the first 30 days** recommended a 7-day post-purchase email nudge as the highest-ROI early intervention.
- Identified regional fulfillment bottlenecks by correlating seller freight duration with delivery times across all 27 states **Roraima and Amapá averaged 29+ day delays vs. São Paulo's 8 days** flagging logistics gaps as a primary churn driver.
- Cleaned and normalized **9 raw CSV files** using Python and pandas; exported to PostgreSQL and computed CRR, RPR, LTV, CSAT, and NPS using **window functions and cohort pivot queries.**
- Built a **4-page interactive Tableau dashboard** covering executive KPIs, geographic revenue breakdown, cohort retention heatmaps, and fulfillment translating raw order data into boardroom-ready insights.

EDUCATION

Vellore Institute of Technology — Bhopal

Bachelor of Technology in Computer Science *(Final Year , CGPA - 7.8)* — 2022–2026

TECHNICAL SKILLS

Languages & Databases: Python (pandas, NumPy, matplotlib, seaborn), SQL (PostgreSQL, MySQL) - Window Functions, CTEs, Cohort Pivot Queries, Query Optimization

Excel & Reporting: Advanced Excel (VLOOKUPs, INDEX/MATCH, Pivot Tables & Charts, Power Query, Conditional Formatting), Executive Report & Slide Preparation

BI & Visualization: Tableau (LOD Expressions, Parameter Actions, Multi-page Dashboard Design, Performance Optimization), Data-Driven Storytelling

Data Processing & Analytics: Exploratory Data Analysis (EDA), ETL Pipeline Development (YAML-driven, config-based), Data Cleaning & Validation, Spatial Clustering, Cohort & Retention Analysis, Trend & Pattern Analysis

Core Competencies: Business Metric Design (CRR, LTV, NPS, CSAT), Large-scale Data Processing (3.1 M+), Hypothesis-Driven Analysis